

# Peer Review Report

## Final Outcome

The problem statement and proposal lacks scope and clarity. The proposal is not fit for purpose from a standpoint of Masters thesis.

## Overview

The project aims to perform a simulation exercise between different types of market participants based on their behavior. The key agents in the simulation are the day trader, a swing trader, a trend trader and a long term investor. All the agents only represent the buy side of the market. The proposal does not mention any correlation among agent behavior and they act independently of each other.

## Strengths

The topic is interesting as well as ambitious.

## Concerns

*I have also made comments in the submitted PDF at the specific places these concerns are.*

1. The proposal does not define a clear quantitative objective. There is absolutely no mention of what metrics would be used to measure the impact of each agent's behavior.
2. The proposal is full of conceptual misunderstandings. The peer makes an assumption that these four agents would move the S&P 500 index. The peer is also confused on what exactly agents will trade, since S&P 500 is an index and not something that is tradeable. Such conceptual flaws are not expected for a Masters Thesis in a Financial Engineering degree.
3. The proposal uses wrong intuitions to stimulate the agent behavior. For example, using a simple moving average strategy crossover does not signify the behavior of a long term investor. There is absolutely no evidence presented in form of a published paper or data to show how different kinds of agents react to different variables.
4. None of the variables that are used to simulate agent behavior are defined. For example the proposal mentions Z score. Is it the Z score of returns? If yes, how do you know that returns are normally distributed?
5. The proposal loosely throws terms such as 'weights' without explaining what they mean.
6. The definition of data, the model and the methodology is completely absent.
7. The end of the paper vaguely mentions non-financial markets without explaining what does that term mean.

## Recommendations

1. The peer is recommended to rework the entire thing. Since behavioral economics is an advanced field of study, a simple literature review of such modelling techniques in financial markets would make up for a good masters thesis.
2. If you indeed want to create something, I think you should do it the following way:
  1. Don't use S&P 500. Model asset prices with Brownian motion having jumps and diffusion. Simulate sufficiently large number of Monte Carlo paths that follow this behavior. You can play with parameters such as drift, jumps and diffusion terms. Or if you are ambitious, you can calibrate these terms directly from S&P 500 historical values.
  2. Define clear and interesting problem statements.
    1. For example, consider two long-only investors. one is a buy-and-hold investor and other who times the market actively that is he only buys when prices are at their local minima. You can define this local minima, the way you want to. One interesting question that I would personally want to explore is **if active market timing statistically outperform a simple buy-and-hold strategy over a large number of Monte Carlo paths.**
    2. Play with coefficients like diffusion and jump and try to answer more interesting questions:
      1. Is active market timing statistically better for a high volatility stock than simple buy-and-hold?
      2. Now also add transaction costs because the guy who is timing the markets is also paying a price for it. Answer the first question again incorporating the transaction costs.

The proposal needs to be re-done from scratch.